

Question 3

```
clearvars
```

```
g_st = readmatrix('q3dat.xlsx')
```

```
g_st = 4x4
    0.8480    -0.2886    -0.4444    -0.5682
    0.5299     0.4619     0.7112     0.9094
         0    -0.8387     0.5446     0.0807
         0         0         0     1.0000
```

```
l0 = 0.5;
l1 = 0.8;
l2 = 0.5;
g_st0 = [eye(3),[0;l1+l2;l0];zeros(1,3),1]
```

```
g_st0 = 4x4
    1.0000         0         0         0
         0     1.0000         0     1.3000
         0         0     1.0000     0.5000
         0         0         0     1.0000
```

```
pb = [0;0;l0];
q2 = [0;l1;l0];
```

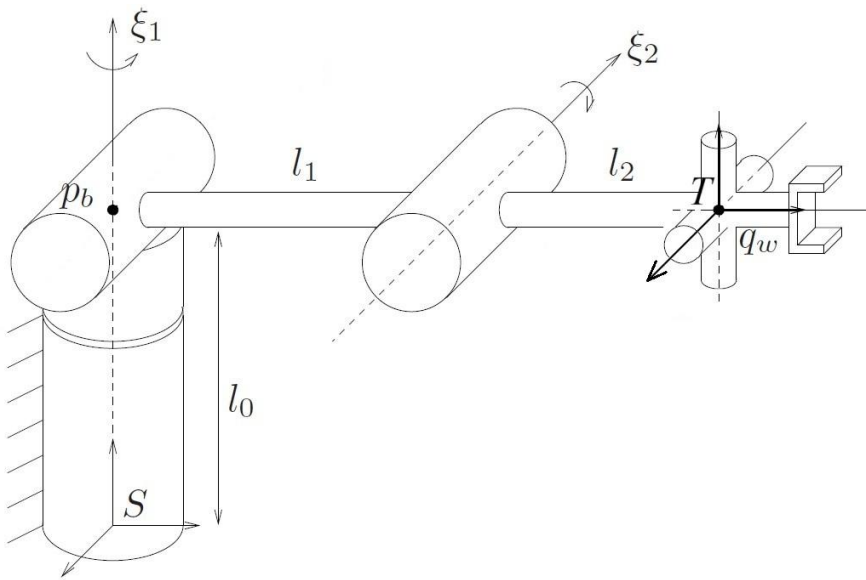
```
omega1 = [0;0;1];
omega2 = [-1;0;0];
```

```
xi1 = [cross(pb,omega1);omega1]
```

```
xi1 = 6x1
     0
     0
     0
     0
     0
     1
```

```
xi2 = [cross(q2,omega2);omega2]
```

```
xi2 = 6x1
     0
    -0.5000
     0.8000
    -1.0000
     0
     0
```



Product of exponentials formula, MLS eq (3.3) p. 87:

$$g_{st}(\theta) = e^{\hat{\xi}_1 \theta_1} e^{\hat{\xi}_2 \theta_2} g_{st}(0) .$$

Therefore,

$$e^{\hat{\xi}_1 \theta_1} e^{\hat{\xi}_2 \theta_2} = g_{st}(\theta) g_{st}(0)^{-1} = g_1 .$$

$$g_1 = g_{st}/g_{st0}$$

$$g_1 = \begin{matrix} 4 \times 4 \\ \begin{matrix} 0.8480 & -0.2886 & -0.4444 & 0.0292 \\ 0.5299 & 0.4619 & 0.7112 & -0.0467 \\ 0 & -0.8387 & 0.5446 & 0.8986 \\ 0 & 0 & 0 & 1.0000 \end{matrix} \end{matrix}$$

If we pick point p on the axis of ξ_2 . Then

$$e^{\hat{\xi}_1 \theta_1} e^{\hat{\xi}_2 \theta_2} p = e^{\hat{\xi}_1 \theta_1} p = g_1 p .$$

From this we can then solve θ_1 using subproblem 1 (MLS p. 99).

```
format long
theta_1 = subproblem_1(xi1,[pb;1],[q2;1],g1*[q2;1])
```

```
theta_1 =
0.558505360638185
```

```
format default
rad2deg(theta_1)
```

```
ans =
32
```

If we now pick any point p . Then

$$e^{\hat{\xi}_2 \theta_2} p = e^{-\hat{\xi}_1 \theta_1} g_1 p .$$

From this we can then solve θ_2 using subproblem 1 (MLS p. 99).

```
pp = g_st0(1:4,4)
```

```
pp = 4×1
      0
      1.3000
      0.5000
      1.0000
```

```
qq = gtwist(xi1,-theta_1)*g1*pp
```

```
qq = 4×1
     -0.0000
      1.0723
      0.0807
      1.0000
```

```
format long
theta_2 = subproblem_1(xi2,[q2;1],pp,qq)
```

```
theta_2 =
      0.994837673636768
```

```
format default
rad2deg(theta_2)
```

```
ans =
      57.0000
```